



NORTHERN WOLVES
AIR CONDITIONING

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Project: **Sunrise Rye Brook - Independent Living / Assisted Living**

Address: **900 King St, Rye Brook, NY 10573**

Mechanical drawings dated: 07.15.2026 (GMP set)

Mechanical Value Engineering Proposals

Northern Wolves reviewed the GMP mechanical set and proposes the value engineering items below. The design today relies on a 5,700 MBH boiler plant with hot water and glycol distribution, hydronic VAV reheat, fan coils and cabinet heaters, plus a full BMS. Most of this can be replaced by the gas rooftop units and heat pump DOAS already in the design and small electric or heat pump terminals, as done on other recent Sunrise communities. Pricing of each item will follow once the owner selects the items to pursue. Pool, spa and sauna systems are by others and not part of this review.

VE	Item	Area	Approvals
VE-01	Heating plant: delete boilers and hot water / glycol systems	AL basement boiler room, chimneys to roof, hot water and glycol piping throughout	Requires VE-02 to VE-06. IMEG redesign; Sunrise approval.
VE-02	VAV reheat: hot water coils to electric reheat	1st floor common areas, IL and AL (VAV-01 to VAV-47)	IMEG redesign and energy code check (VAV minimums); electrical service capacity by EE.
VE-03	Common areas: single-zone rooftop units in lieu of VAV zoning (alternate to VE-02)	1st floor dining, lobby, activity, kitchen and offices	IMEG zoning concept, ventilation air source and redesign; Sunrise approval.
VE-04	Cabinet heaters: hot water to electric	Entries, vestibules and stairs (CH-1 to CH-12)	IMEG; EE.
VE-05	Basement back-of-house fan coil units	IL and AL basement: storage, trash, laundry, electrical, mechanical, staff rooms (FCU-A)	IMEG; fire protection designer to confirm freeze-protection locations.
VE-06	Garage ramp snow melt	North and south garage ramps (HX-3, P-3 / P-4, 120 F glycol)	Owner decision; civil / landscape for drainage.
VE-07	Fire / smoke dampers at apartment duct branches	Floors 1 to 4, IL and AL corridors (about 360 FSD tags)	IMEG / HKS code analysis (AL Group I-1 smoke barriers), building department.
VE-08	Apartment bath exhaust to the DOAS energy recovery	All apartments, IL and AL (TX-A)	IMEG; DOAS exhaust airflow check.
VE-09	Controls: standalone equipment controls in lieu of BMS	Building-wide	Sunrise approval of the controls standard.

VE-01 requires VE-02 (or VE-03) and VE-04 to VE-06. Electrical work for electric heat by EC; electrical service capacity to be confirmed by the EE and utility.

VE-01 Heating plant: delete boilers and hot water / glycol systems

Area: AL basement boiler room, chimneys to roof, hot water and glycol piping throughout

As designed

- Three 1,999 MBH gas boilers B-1, B-2, B-3 (5,700 MBH output), primary / secondary with HS-1
- Pumps BP-1/2, PP-1/2, P-1 to P-4, expansion tanks ET-1/2, air separator AS-1, heat exchanger HX-3 to 30% glycol
- Fire-rated chimney enclosures basement to roof, combustion air areaway, water treatment, hot water and glycol distribution to the 1st floor and basement
- Connected heating load about 2,430 MBH: the plant is about 2.3 times the load

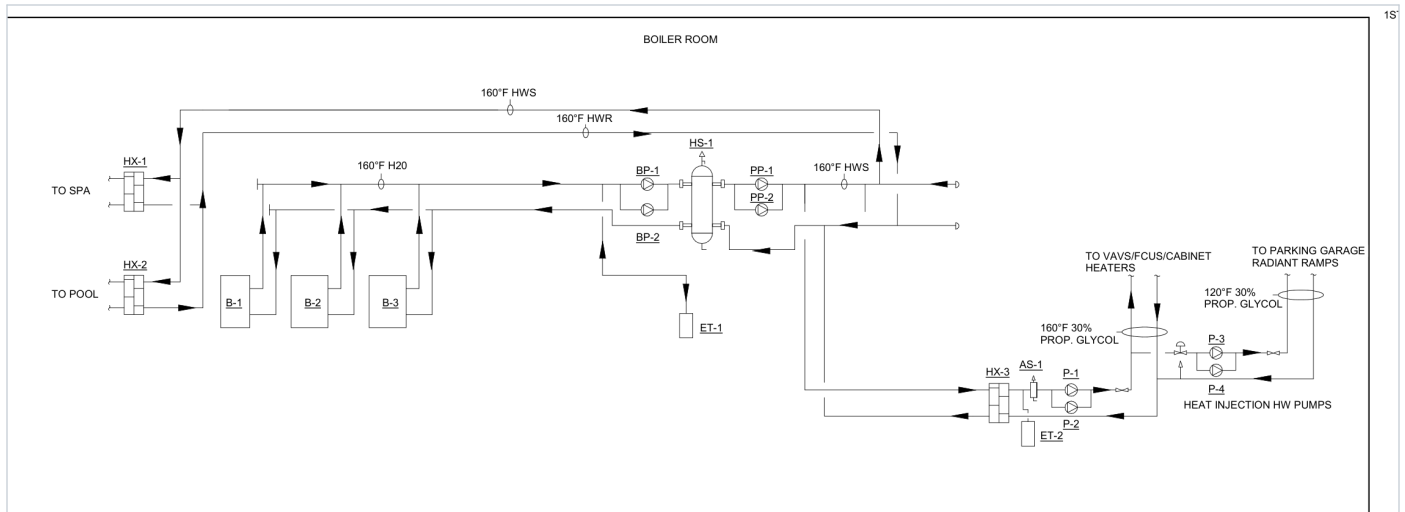
Proposed

- Delete the boiler plant and all hot water / glycol distribution; the building runs on gas rooftop units, heat pump DOAS and the electric terminal heat in VE-02 to VE-06
- No boiler room equipment, chimneys, combustion air, water treatment, flushing or hydronic balancing
- Pool and spa heating by the pool contractor (by others)

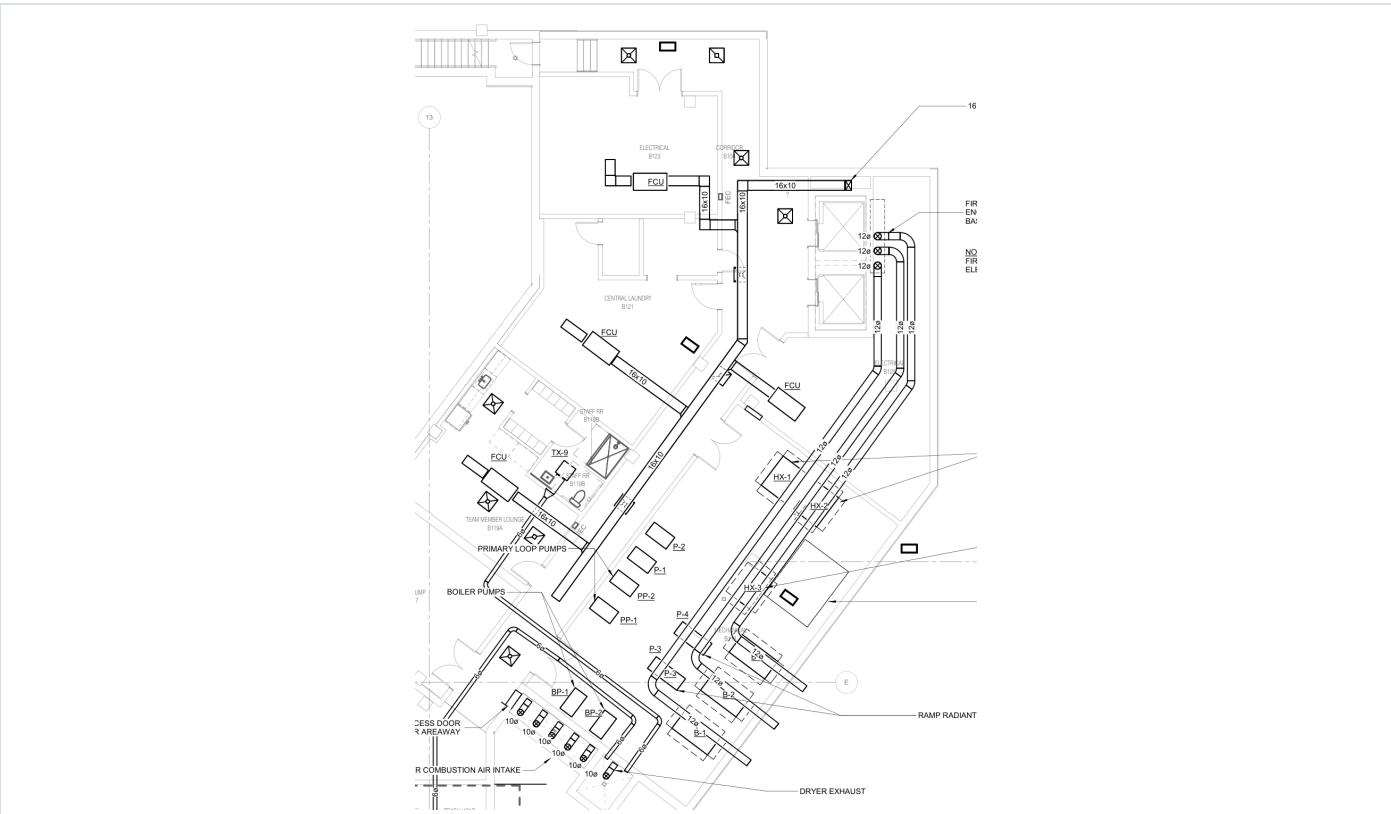
Benefits: Largest single reduction in mechanical, controls and GC scope (chimney shafts, areaway, boiler room fit-out). Removes the boiler plant maintenance and its BMS sequences.

Electrical impact: See VE-02 to VE-05

Approvals / coordination: Requires VE-02 to VE-06. IMEG redesign; Sunrise approval.



M5000 Hydronic flow diagram - boiler room



M2002 AL basement - boiler room, pumps, heat exchangers, chimney enclosures

BOILER SCHEDULE - HOT WATER (BLR)										
NOTES: 1.										
TAG NAME	SERVICE	FUEL	MBH	MAXIMUM DIMENSIONS			WEIGHT LBS	MANUFACTURER	MODEL	NOTES
		TYPE	INPUT	OUTPUT	WIDTH	HEIGHT	DEPTH	DRY		
B-1	BUILDING HEATING	FIRE TUBE	1999	1800	3'-0"	6'-8"	6'-2"	1,750	LAARS MAGNITHERM FT	2000
B-2	BUILDING HEATING	FIRE TUBE	1999	1800	3'-0"	6'-8"	6'-2"	1,750	LAARS MAGNITHERM FT	2000
B-3	BUILDING HEATING	FIRE TUBE	1999	1800	3'-0"	6'-8"	6'-2"	1,750	LAARS MAGNITHERM FT	2000

PUMP SCHEDULE (P)							
NOTES: 1.							
TAG NAME	LOCATION	SERVICE	GPM DESIGN	FT. HEAD	WEIGHT LBS	OPERATING	MANUFACTURER AND MODEL
BP-1	AL BASEMENT	BOILER	381	30	141		ARMSTRONG 4300 0610-005
BP-2	AL BASEMENT	BOILER	381	30	141		ARMSTRONG 4300 0610-005
P-1	AL BASEMENT	HX-3	55	40	67		ARMSTRONG 4300 1205-001.5
P-2	AL BASEMENT	HX-3	55	40	67		ARMSTRONG 4300 1205-001.5
P-3	AL BASEMENT	HX-3	56	30	67		ARMSTRONG 4300 1205-000.7
P-4	AL BASEMENT	HX-3	56	30	67		ARMSTRONG 4300 1205-000.7
PP-1	AL BASEMENT	BOILER	0	0	0		

M6001 Boiler and pump schedules

VE-02 VAV reheat: hot water coils to electric reheat

Area: 1st floor common areas, IL and AL (VAV-01 to VAV-47)

As designed

- 47 VAV boxes with hot water reheat coils (629 MBH, 45 GPM) on 5 rooftop units
- Hot water piping, control valves, balancing and insulation to every box (M2014 to M2016)

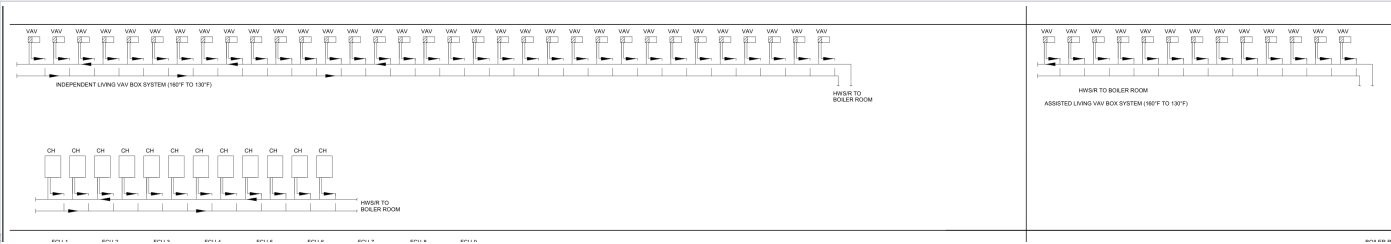
Proposed

- Factory electric reheat coils with SCR control on the same VAV boxes; no piping, no control valves
- IMEG to confirm which interior zones need reheat at all (rooftop units already supply gas-heated air); cooling-only boxes where reheat is not required

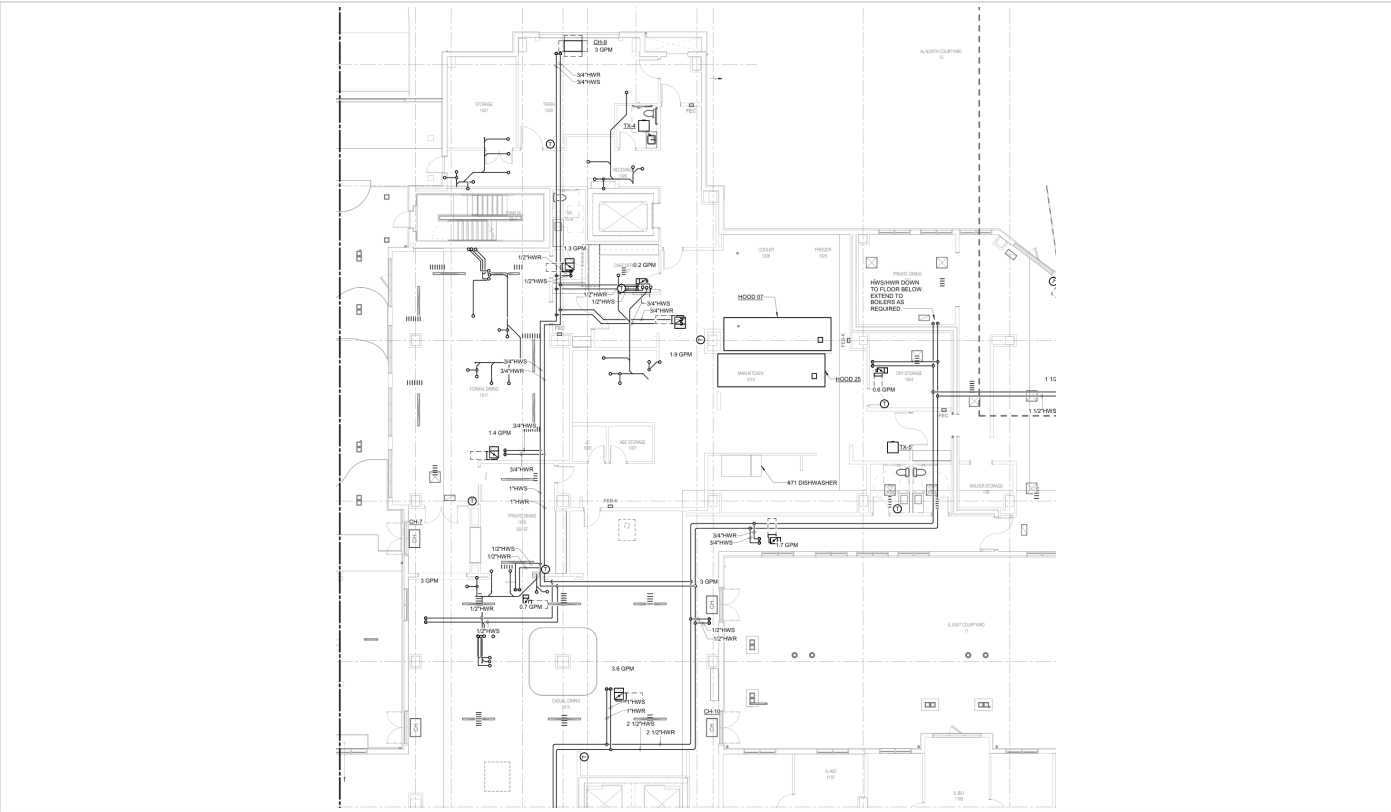
Benefits: Deletes 1st floor hot water piping and 47 valve / coil packages; fewer BMS points.

Electrical impact: About +185 kW (less if interior zones are cooling-only)

Approvals / coordination: IMEG redesign and energy code check (VAV minimums); electrical service capacity by EE.



M5000 Hydronic flow diagram - IL and AL VAV reheat and cabinet heaters



M2015 1st floor piping IL area B - hot water to VAV boxes and cabinet heaters

VAV SCHEDULE - HYDRONIC REHEAT																							
TAG	AHU	ROOM	MODEL	SIZE		CFM	STATIC PRESSURE			NC LEVELS			HOT WATER HEATING COIL							UNIT INFO			
				UNIT	OUTLET		MAX	MIN	INLET	DOWN	MIN	SAD	DS	CFM	MAX	EAT	ENT	LP	APF		QTY	RW	PRF
VAV-01			DESV	12	16x15	1200	625	1	0.25	0.24	20	24	825	20.9	60	180	90	0.23	1.5	122.5	0.24	10	RH
VAV-02			DESV	14	20x17.5	1450	1350	1	0.25	0.19	20	19	1350	44	60	180	90	0.17	2.4	122.6	0.3	10	RH
VAV-03			DESV	14	20x17.5	2250	1150	1	0.25	0.38	23	22	1150	37.5	60	180	90	0.35	1.9	120	0.22	10	RH
VAV-04			DESV	08	12x10	730	375	1	0.25	0.32	22	28	375	12.2	60	180	90	0.3	0.8	127.1	0.15	10	RH
VAV-05			DESV	08	12x10	430	400	1	0.25	0.13	17	25	400	13	60	180	90	0.12	0.8	127.6	0.17	10	RH
VAV-06			DESV	08	12x10	430	400	1	0.25	0.13	17	25	400	13	60	180	90	0.12	0.8	127.6	0.17	10	RH
VAV-07			DESV	10	14x12.5	730	250	1	0.25	0.08	23	27	250	8.1	60	180	90	0.08	1	143.9	0.19	10	RH
VAV-08			DESV	04	12x8	290	100	1	0.25	0.17	27	34	100	3.3	60	180	90	0.07	0.89	144.6	0.1	10	RH
VAV-09			DESV	08	12x10	390	150	1	0.25	0.06	17	25	150	4.9	60	180	90	0.05	0.5	127.6	0.19	10	RH
VAV-10			DESV	06	12x8	425	250	1	0.25	0.36	23	24	250	8.9	60	180	90	0.21	0.7	124.3	0.1	10	RH
VAV-11			DESV	10	14x12.5	590	500	1	0.25	0.12	20	24	500	16.8	60	180	90	0.11	1.1	127.6	0.14	10	RH
VAV-12			DESV	14	20x17.5	1350	1350	1	0.25	0.15	18	19	1350	42.3	60	180	90	0.14	2.3	122.3	0.28	10	RH
VAV-13			DESV	10	14x12.5	630	600	1	0.25	0.14	20	24	600	19.5	60	180	90	0.13	1.3	128.7	0.18	10	RH
VAV-14			RFPRV	15	18x17.5	400	400	1	0.24	0.14	35	34	400	19.4	60	180	90	0.15	1.1	128.9	0.18	10	RH

M6000 VAV schedule - hydronic reheat (partial)

VE-03 Common areas: single-zone rooftop units in lieu of VAV zoning (alternate to VE-02)

Area: 1st floor dining, lobby, activity, kitchen and offices

As designed

- 5 large VAV rooftop units (7,000 to 11,000 CFM) serving 47 VAV zones with reheat

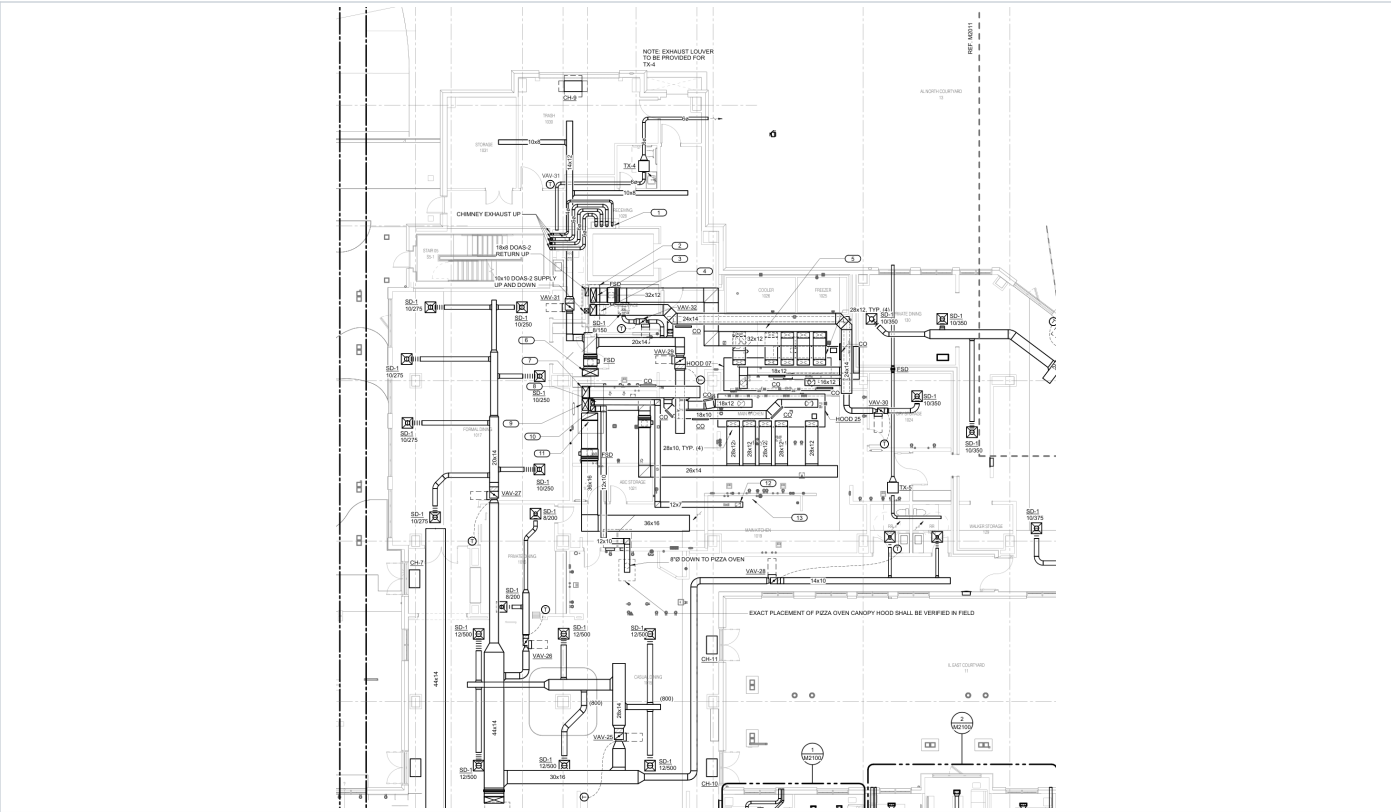
Proposed

- Large open spaces on single-zone constant-volume gas / DX rooftop units, one per space type
- Offices and small rooms on ductless heat pumps; delete the VAV boxes and reheat
- Ductless units recirculate only: each small room keeps a ventilation air branch (code outdoor air) with a diffuser, preferably tempered air from the DOAS (or a small ERV per room cluster), not raw rooftop unit supply air that would fight the ductless unit
- Same approach as other recent Sunrise communities (e.g. Sunrise of Northport: small single-zone rooftop units plus heat pump terminals, no VAV, no boilers)

Benefits: Deletes 47 VAV boxes, reheat and their controls; simpler operation for the community.

Electrical impact: Heat pumps for small rooms only (less than electric reheat)

Approvals / coordination: IMEG zoning concept, ventilation air source and redesign; Sunrise approval.



M2011 1st floor ductwork - VAV boxes and cabinet heaters in dining / kitchen area

VE-04 Cabinet heaters: hot water to electric

Area: Entries, vestibules and stairs (CH-1 to CH-12)

As designed

- 12 hot water cabinet heaters (234 MBH) with piping and BMS control

Proposed

- Electric cabinet heaters with integral thermostats, no piping, no BMS points

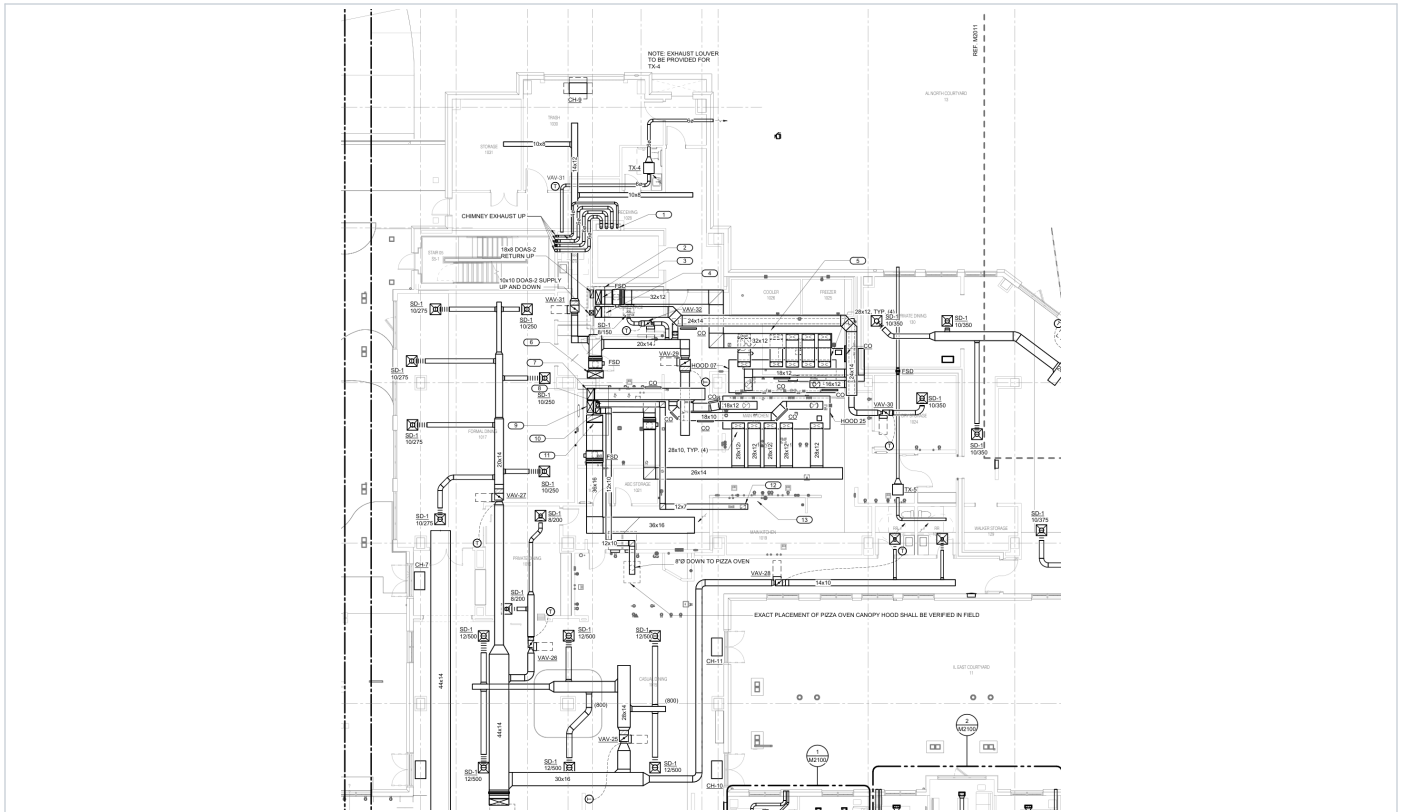
Benefits: Deletes piping to 12 remote locations; standalone control.

Electrical impact: About +70 kW connected

Approvals / coordination: IMEG; EE.

CABINET HEATER SCHEDULE																			
NOTES: 1.COORDINATE COLOR SELECTION WITH ARCHITECT.																			
TAG NAME	LOCATION		DIMENSIONS			FAN		COIL	GPM	COIL		ELECTRIC		MAXIMUM DIMENSIONS			WEIGHT		MO
	NAME	SERVICE	WIDTH	HEIGHT	DEPTH	CFM DESIGN	DRIVE			MBH	VOLTS	PH		WIDTH	HEIGHT	DEPTH	DRY	MANUFACTURER	
CH-1		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105		MODINE	CW0
CH-3		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105		MODINE	CW0
CH-4		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105		MODINE	CW0
CH-5		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105		MODINE	CW0
CH-6		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105		MODINE	CW0
CH-7		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105		MODINE	CW0
CH-8		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105		MODINE	CW0
CH-9		BOILER	40"	12"	24"	1,050	CENTRIFUGAL	HOT WATER	5.8	35552	120	1	5'-11 3/4"	2'-4"	1'-0"	185		MODINE	CW0
CH-10		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105		MODINE	CW0
CH-11		BOILER	40"	12"	24"	450	CENTRIFUGAL	HOT WATER	3	18062	120	1	4'-0 3/4"	2'-1"	9 3/4"	105		MODINE	CW0

M6001 Cabinet heater schedule



M2011 1st floor - CH-7, CH-9, CH-10, CH-11 locations

VE-05 Basement back-of-house fan coil units

Area: IL and AL basement: storage, trash, laundry, electrical, mechanical, staff rooms (FCU-A)

As designed

- FCU-A, quantity 20, "basement heating", hot water / glycol heating coil only (schedule data blank); M5000 shows FCU-1 to FCU-17 on the glycol loop with RAD-1/2
- Each FCU takes a small DOAS branch at its inlet and recirculates the room (booster fan plus reheat)
- No glycol or hot water piping to these units is drawn on M2001 / M2003

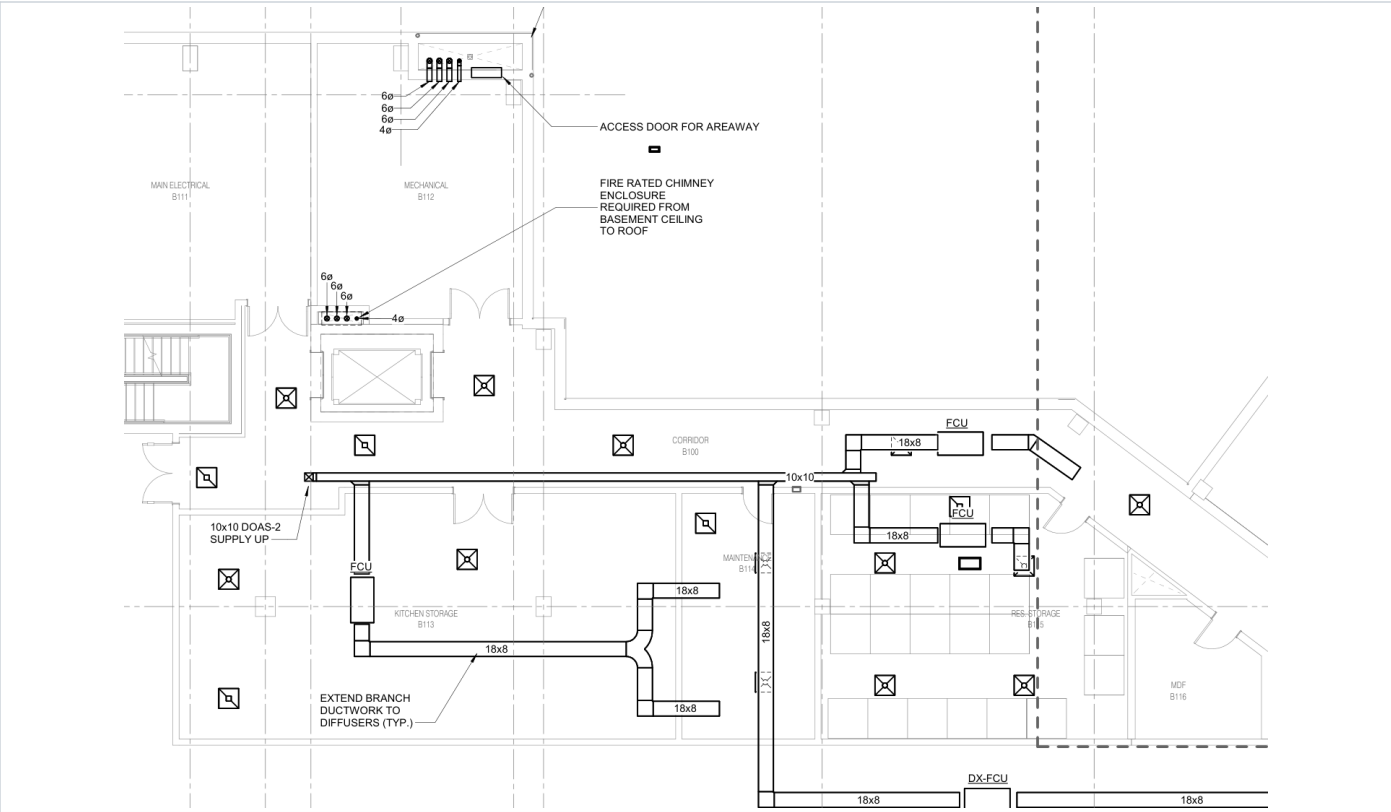
Proposed

- Delete the 20 FCUs and RAD-1/2; extend the DOAS branch ducts directly to the room diffusers
- Electric unit / wall heaters with integral thermostats only where there is freeze risk (trash rooms, rooms at areaways, louvers or exterior walls)
- Occupied rooms (staff lounge, corridor) on a small ductless heat pump or duct heater
- Electrical rooms, MDF and laundry need cooling and ventilation, not heat: ductless AC for MDF and electrical rooms, thermostatic exhaust / make-up for laundry (not on the drawings today)

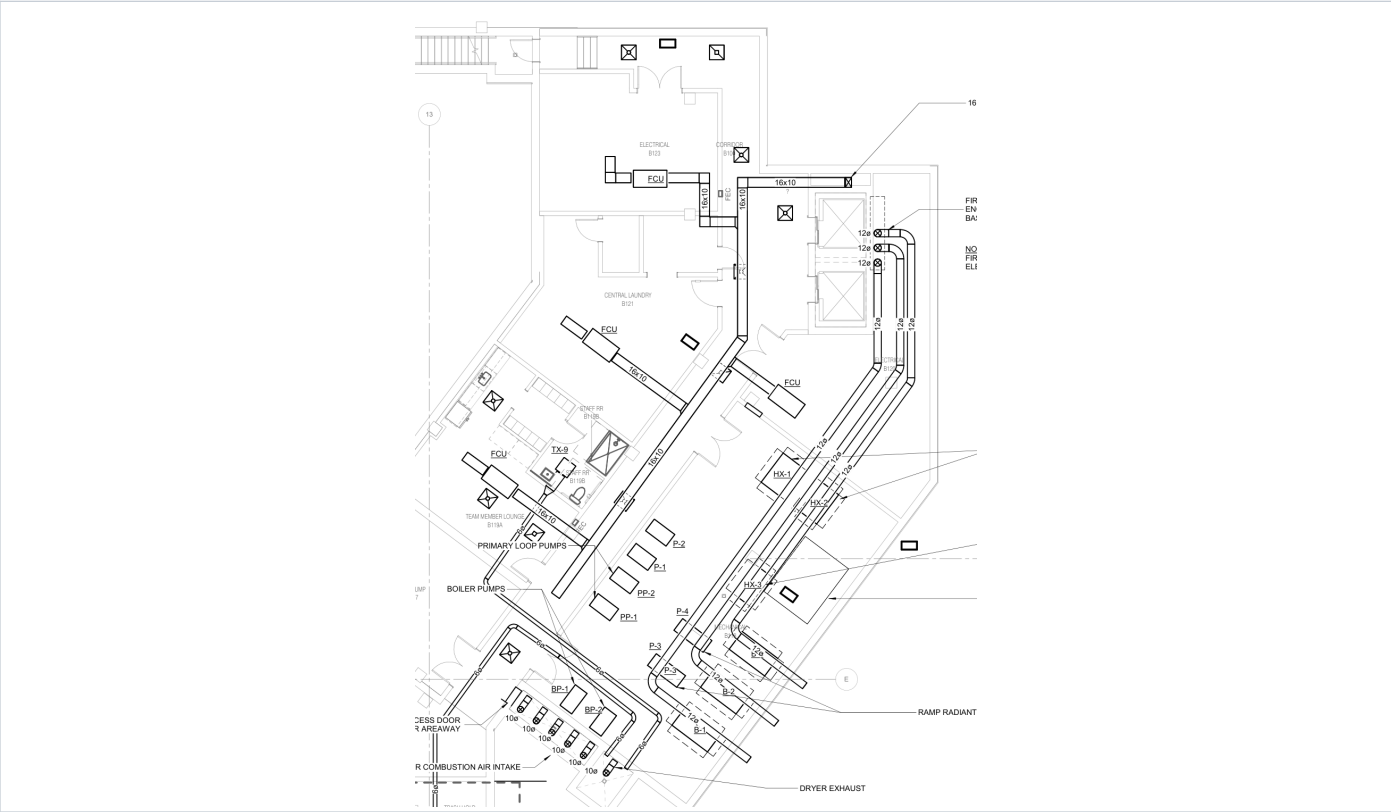
Benefits: Deletes 20 fan coils, glycol piping and controls. Corrects the missing cooling in equipment rooms.

Electrical impact: About +30 kW connected

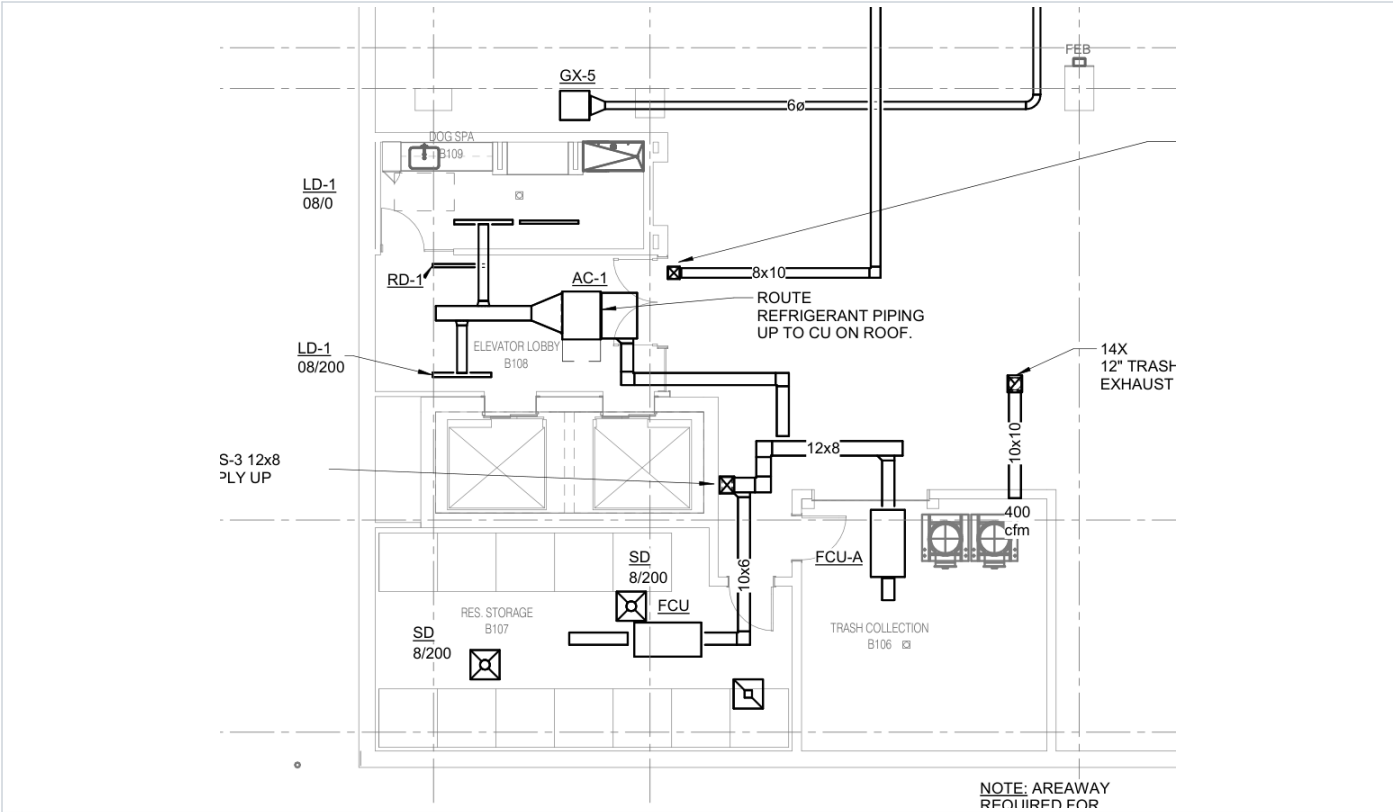
Approvals / coordination: IMEG; fire protection designer to confirm freeze-protection locations.



M2002 AL basement - FCUs in kitchen storage, residents' storage, corridor



M2002 AL basement - FCUs in electrical B123, central laundry, staff lounge



M2000 IL basement - FCU in residents' storage, FCU-A in trash collection

FAN COIL UNIT SCHEDULE - HYDRONIC (FCU)																						
NOTES: 1.																						
TAG NAME	QUANTITY	SERVICE	TYPE	FAN			HEATING COIL							ELECTRICAL		CONTROL		MAXIMUM DIMENSIONS			WEIGHT LBS	MANUF
				CFM DESIGN	EXT S.P. IN. WC	MOTOR MHP [E]	MBH	EAT DB °F	LAT DB °F	GPM DESIGN	W.P.D. FT. HD.	EWI °F	LWT °F	VOLTS	PH	TYPE [H]	DIAGRAM	WIDTH	HEIGHT	DEPTH	DRY	
FCU-A	20	BASEMENT HEATING	HORIZONTAL	0	0	0	0	0	0	0	0	0	0	0	1			0"	0"	0"	0	

M6000 Fan coil unit schedule (FCU-A, quantity 20, no data)

VE-06 Garage ramp snow melt

Area: North and south garage ramps (HX-3, P-3 / P-4, 120 F glycol)

As designed

- In-floor radiant snow melt loops on both ramps from the boiler plant through HX-3

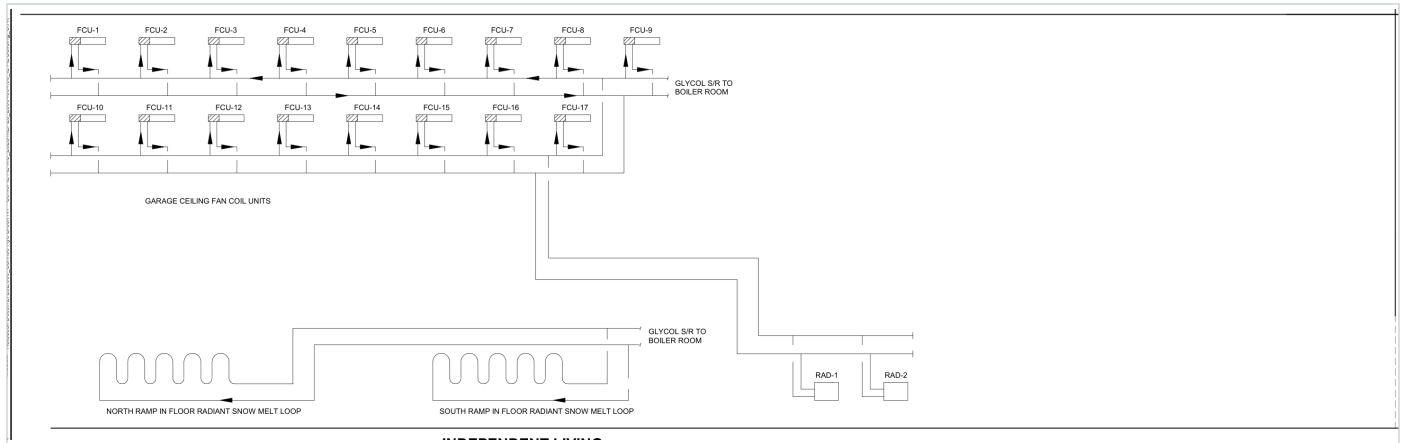
Proposed

- Option A: no snow melt; trench drains and salting (owner decision)
- Option B: electric snow melt cable by EC, storm-only operation with load shedding

Benefits: Removes the long glycol mains from the boiler room to the ramps and the snow melt pumps.

Electrical impact: Option A: 0; option B: about +240 kW (storm only)

Approvals / coordination: Owner decision; civil / landscape for drainage.



M5000 Hydronic flow diagram - basement FCUs, RAD-1/2 and ramp snow melt loops

VE-07 Fire / smoke dampers at apartment duct branches

Area: Floors 1 to 4, IL and AL corridors (about 360 FSD tags)

As designed

- Combination fire / smoke damper at every DOAS supply / return branch into each apartment (M2023 note: "provide fire/smoke damper at all DOAS supply/return penetrations through demising walls")
- Each damper needs power, fire alarm interface, access door in a finished ceiling and periodic testing

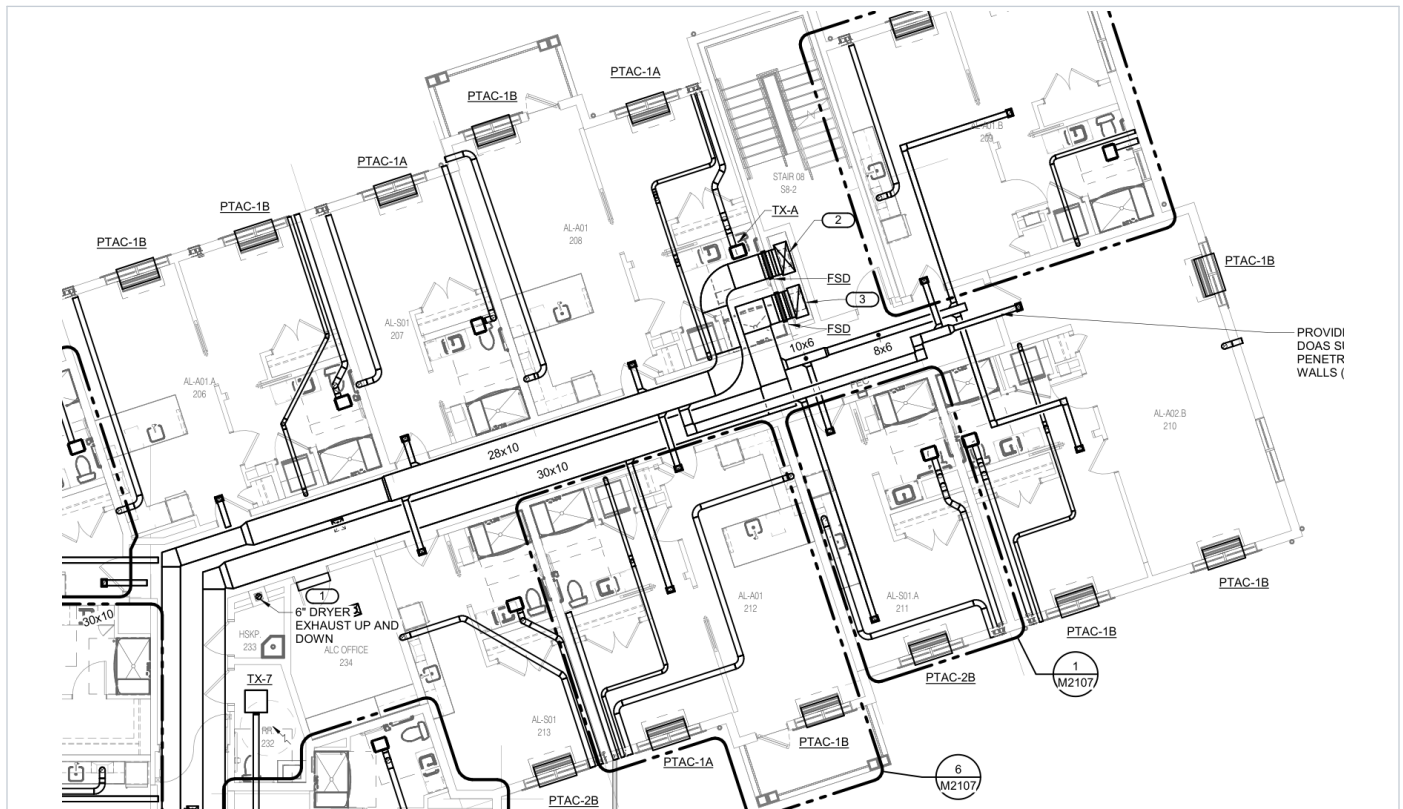
Proposed

- Apartment branches (6x4 to 8x6) under 100 sq in, 26 ga steel, no openings into the corridor, above the ceiling, with 12" steel sleeve: no damper required at the corridor fire partition (IBC 717.5.4 exception); plain fire dampers as fallback
- Fire / smoke dampers kept at shafts and smoke barriers

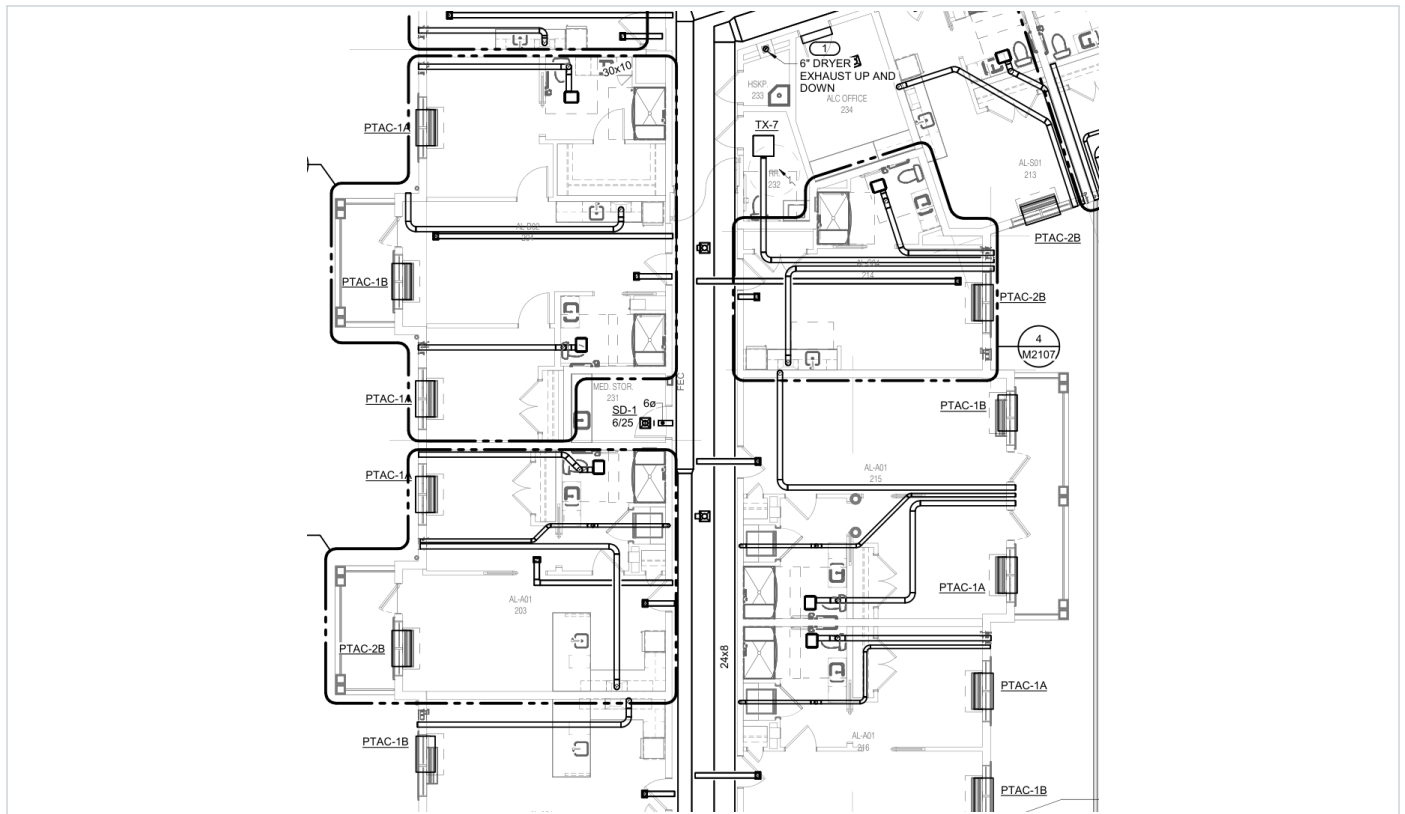
Benefits: Removes about 283 fire / smoke dampers with their power, fire alarm modules and access doors (mechanical, electrical and fire alarm scope).

Electrical impact: Reduces damper power circuits

Approvals / coordination: IMEG / HKS code analysis (AL Group I-1 smoke barriers), building department.



M2023 AL 2nd floor - DOAS corridor ducts, FSD at shaft and note for all units



M2023 AL 2nd floor - DOAS branches with FSD into each apartment

VE-08 Apartment bath exhaust to the DOAS energy recovery

Area: All apartments, IL and AL (TX-A)

As designed

- Each apartment bath has its own exhaust fan TX-A (75 CFM) ducted out the exterior wall
- DOAS return is taken from corridors and a few rooms; some AL unit types show a DOAS return grille RG-2, the floor plans connect the return trunk to only a few (to be clarified)

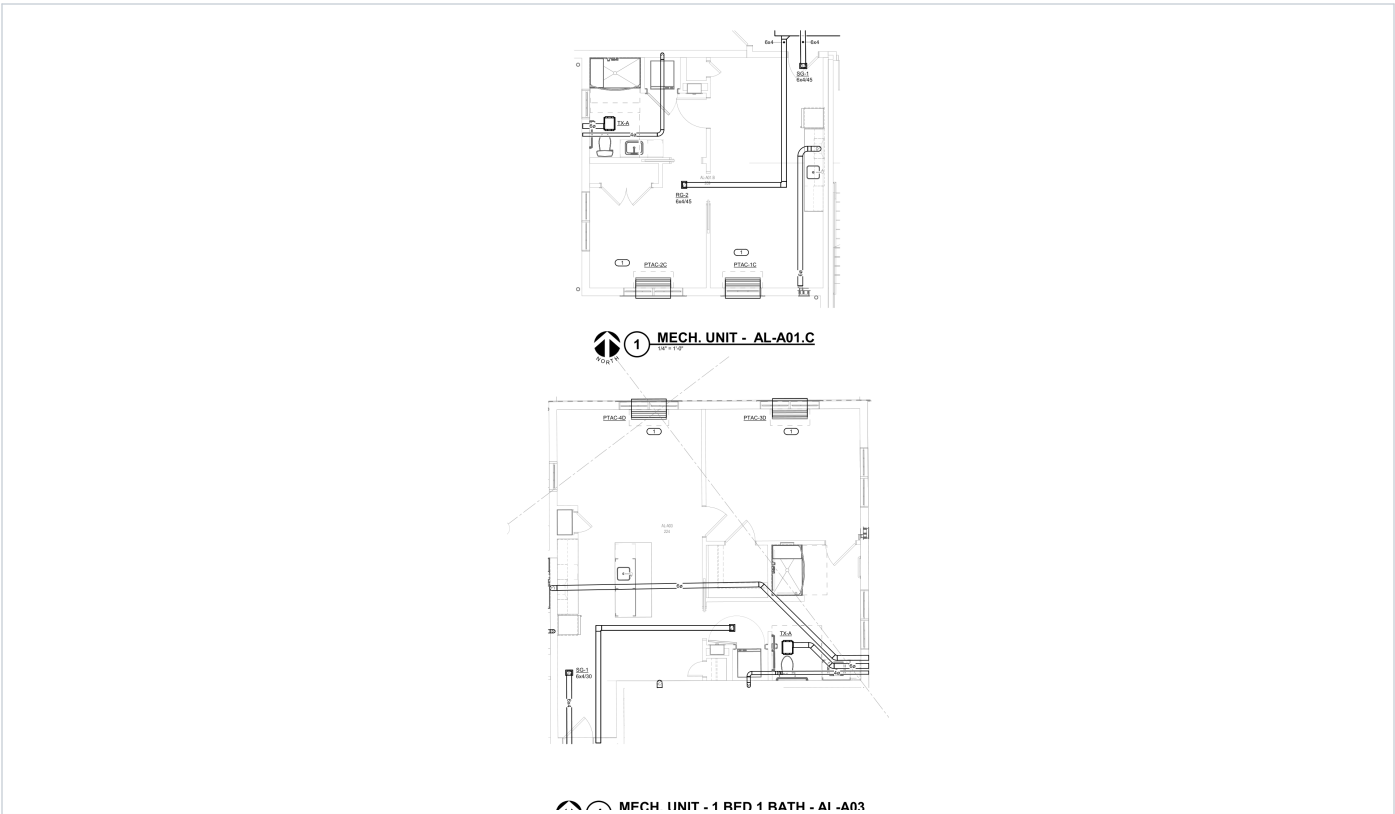
Proposed

- Duct apartment bath exhaust to the DOAS exhaust trunk in the corridor (as on Sunrise of Northport); the DOAS energy wheel recovers the exhaust heat
- Delete the individual TX-A fans, wall caps and their electrical circuits; delete RG-2 in the units

Benefits: Fewer wall penetrations and fans to maintain; better energy recovery and code compliance; the corridor return trunk serves every apartment.

Electrical impact: About -11 kW (fans deleted)

Approvals / coordination: IMEG; DOAS exhaust airflow check.



M2108 AL unit plans - TX-A bath exhaust to exterior, SG-1 / RG-2 DOAS grilles

VE-09 Controls: standalone equipment controls in lieu of BMS

Area: Building-wide

As designed

- Full BMS by the BMS contractor for boilers, pumps, heat exchangers, glycol, snow melt, VAV reheat, FCUs, cabinet heaters, rooftop units and DOAS

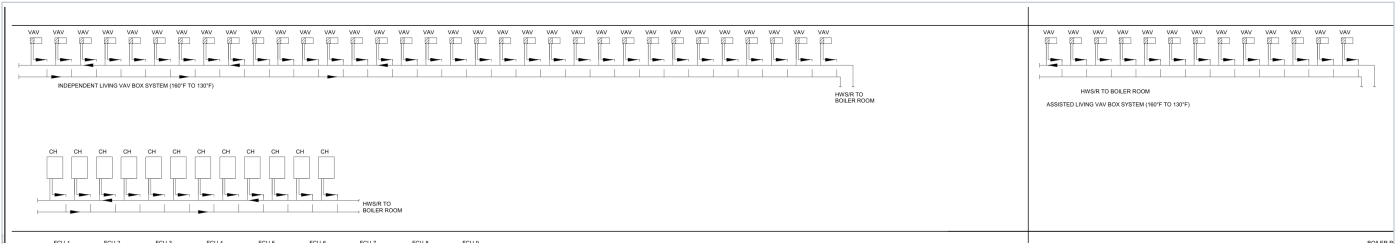
Proposed

- With VE-01 to VE-06 the hydronic sequences (about 120 to 150 points) are gone
- Rooftop units and DOAS on their factory controls with network cards for future monitoring; heaters, mini-splits and PTACs on standalone thermostats (as on other Sunrise communities)

Benefits: Large reduction or deletion of the BMS contract; simpler operation for the community staff.

Electrical impact: -

Approvals / coordination: Sunrise approval of the controls standard.



M5000 - BMS-controlled hydronic terminals eliminated by VE-01 to VE-05

Coordination items found during the review (RFI to follow)

- Basement piping sheets M2001 / M2003 show no boiler room, main or glycol piping.
- FCU counts do not match: schedule FCU-A quantity 20 (all data blank), M5000 FCU-1 to FCU-17, plans about 12.
- Pump schedule PP-1 / PP-2 blank. RTU-5 heating output 864 MBH is greater than its 600 MBH input.
- Electrical rooms, MDF and central laundry have no cooling or ventilation scheduled (heating-only FCUs).
- AL unit plans show DOAS return grilles RG-2 in some unit types; floor plans connect the return trunk to only a few rooms.